FCOM gap fixes (program overview)

Overview of the 207 own commits of the FCOM gap program: A380X and A32NX alerts, displays, FADEC, FMS predictions and VNAV, GPWS / TAWS predictive terrain and obstacles, flight control laws and system models, grouped in feature families with how to test each

Aircraft	A32NX and A380X
Where to find it	Everywhere in the cockpit: E/WD and SD alerts, PFD / ND / MFD / MCDU pages, overhead panels, flyPad Failures
Branch	`program/fcom-gap-fixes` : a fork-only program branch cut from develop `9ad68d612` (it needs develop; it is not an upstream pull request branch), with FlyByWire master 66021c244 merged in. The same changes were placed on single-topic `core/*` branches where their base allows it; this manual describes them as one program. The 207 commits are listed by family in the commit list of the branch.
Status	Fork-only program. Mixed: the 2026-10-12 to 2026-10-14 gap batches were committed without sim tests; the later FADEC, VNAV, A32NX / A380X ECAM, ELAC, A380X F/CTL and GPWS / TAWS work was sim-tested in install groups (per family below). 15 of the commits are `test(...)` , `docs` and `chore` commits (spec conversions, flaky test fixes, README files for the TAWS grid generators, CHANGELOG housekeeping): no user-visible change.
Pictures	Pictures only in the GPWS / TAWS family (in-sim screenshots of the ND TERR AHEAD message and threat areas, taken during development); all other families are described in text.

1. A380X FWS alert units by ATA chapter
2. A32NX FWC alerts, SD and PFD indications
3. A380X SD, PFD, EWD and panel display rules
4. FADEC, EIU, engine idle and relight
5. FMS predictions and VNAV
6. A32NX GPWS: modes 1 and 2 rewritten, terrain ahead alerting (TAD), obstacle alerts and terrain clearance floor (TCF)
7. Surveillance: GPWS test, TCAS traffic and failures, windshear and radio height voices, ILS side
8. Flight control law fixes (ELAC, A380X protections)
9. Aircraft system models: hydraulics, bleed, fuel, electrics, brakes, APU, ice, pressurization, ADIRS
10. Panels, communications, effects, flyPad and GSX

1. A380X FWS alert units by ATA chapter

> **Aircraft:** A380X | **Where:** E/WD, STATUS page, SD pages, overhead panels, flyPad Failures | **Status:** The alerts of 2026-10-28 (avionics ventilation, pack and mixer, hydraulic heat exchanger, CDS DU NOT MONITORED and mailbox access) were sim-tested by the user (DU NOT MONITORED both loops, CAPT / F/O MAILBOX ACCESS FAULT, HYD HSMU, fuel heat exchanger valve, heat exchanger air leak and detection, engine pump overheat): OK. The 2026-10-12 to 2026-10-14 alerts (door, ELEC busbar, landing gear, engine, hydraulic, CAB PRESS, COM, datalink, fuel) were not sim-tested.

What it is for

The A380X flight warning system (FWS) lacked many alerts of the FCOM. They were added unit by unit, by ATA chapter, each with its trigger, flight phase inhibitions, procedure lines, STATUS lines and, where the alert had no way to be triggered, a new flyPad failure: air conditioning (pack valve, pack overheat, pack regulation, mixer, ram air, cabin pressure sensors), avionics ventilation, cabin pressure, doors and T.O CONFIG for every door, ELEC busbars, landing gear and NAV disagree, engines (levels, THR LEVER ABOVE IDLE, attention box, tire pressure), hydraulic pressure switches and heat exchangers, fuel, CDS and mailbox access, datalink and COM, CIDS, APU. It also adds the unit-feet radio height callouts, the PITCH PITCH and COMPANY MESSAGE voices and a fix that keeps the FWS sound manager from looping when the audio functions are lost.

How to use

1. Nothing to set: the alerts appear on the E/WD when their condition occurs.
2. To provoke one: flyPad > Failures, choose the failure named in the test below.

How to test

#	Do this	Expected result
1	A380X powered on the ground or in cruise. flyPad > Failures > ATA 21 > **Pack 1 overheat**, Fail.	AIR PACK 1 OVHT on the E/WD with its procedure; Restore clears it.
2	flyPad > Failures > ATA 21 > **Avionics ventilation left blowing filter clogged**, then the right one.	After both: VENT AVNCS BLOWING DEGRADED on the E/WD (both circuits on their backup valves).
3	flyPad > Failures > ATA 79 > **Engine 1 oil filter clog**.	ENG 1 OIL FILTER CLOGGED comes without chime and without MASTER CAUT (FCOM level).
4	flyPad > Failures > ATA 23 > **CIDS 1 (cabin intercommunication data system)**, then CIDS 2 and 3.	With all three failed: CAB COM CIDS 1+2+3 FAULT.

2. A32NX FWC alerts, SD and PFD indications

> **Aircraft:** A32NX | **Where:** E/WD, STATUS, SD pages, PFD, ND | **Status:** Sim-tested by the user in the coverage program groups G2 / G3 (A32NX ECAM: NAV, F/CTL, mixed alerts, all 6 cards passed, 2026-10-20). The 2026-10-12 to 2026-10-14 alerts (AIR bleed, HYD, ELEC, fuel, doors, oxygen, STATUS paging, SD pages) were not sim-tested.

What it is for

The A32NX flight warning computer got the missing FCOM cautions with their procedure and STATUS lines: AIR bleed cautions, HYD cautions and warnings, ELEC bus, ESS bus and TR alerts, F/CTL surface cautions (both ELACs failed show one ELAC 1+2 FAULT), wing anti-ice, L/G DOORS NOT CLOSED, ventilation, COM, WHEEL HYD SEL FAULT, SIDESTICK and FAC alerts. The SD pages (BLEED, HYD, ELEC, DOOR/OXY, WHEEL, APU) show amber where the FCOM says so; the STATUS page now pages instead of cutting after 16 lines; the PFD gets V1 INOP and ILS / DME flags and the FMA line 3 messages. Pneumatic, electrical and brake control failures were added to the flyPad.

How to use

1. Nothing to set. Provoke an alert with flyPad > Failures.

How to test

#	Do this	Expected result
1	A32NX in cruise. flyPad > Failures > ATA 27, fail **ELAC 1** then **ELAC 2**.	One alert ELAC 1+2 FAULT (not two ELAC FAULT alerts) with its procedure.
2	flyPad > Failures > ATA 36 > **Engine 1 bleed valve stuck closed**, with the engine running and the bleed on.	ENG 1 BLEED FAULT caution with its procedure; the SD BLEED page shows the bleed valve amber when it is not in the commanded position.
3	Cabin altitude above 14,000 ft (rapid decompression failure, flyPad > Failures > ATA 21).	The passenger oxygen masks drop by themselves.
4	Make a STATUS page longer than 16 lines per column (several failures).	A green arrow shows that more follows; CLR shows the next page.

3. A380X SD, PFD, EWD and panel display rules

> **Aircraft:** A380X | **Where:** SD ELEC / F/CTL / WHEEL / BLEED pages, PFD, EWD, glareshield, KCCU | **Status:** Committed in the FCOM gap program of 2026-10-12 to 2026-10-14 (batches P1 to P20 and Tier 2) without sim tests, on the decision that the community tests at pull request time. Unit tests only.

What it is for

Display rules of the A380 FCOM that were wrong or missing: SD ELEC AC / DC (galley and commercial shed legends, generator DISC and OVHT, APU BAT voltage), F/CTL page (BKUP CTL ACTIVE), WHEEL page (WHEEL MORE, BCS / EBCU / SCS lines, brake arcs), automatic SD pages (CRUISE from 1,500 ft after take-off), PFD radio height (median of three RAs, same on both PFDs), the CAPT PFD reading the CAPT side FMS data, FMA messages (MAN DTO for a derated take-off, SPD BRK, CHECK APPR SEL), the VD radio altitude zone, the OANS destination arrow, the AGENT SQUIB light and the loudspeaker volume knobs, and the hosted display gates (a hidden gauge no longer takes the mouse of the DU).

How to use

1. Nothing to set.

How to test

#	Do this	Expected result
1	A380X, take off and climb through 1,500 ft.	The SD shows the CRUISE page from 1,500 ft after take-off (not 60 s after the flaps are up).
2	Lose all PRIMs and SECs (flyPad > Failures > ATA 27) and look at the SD F/CTL page.	BKUP CTL ACTIVE in amber.
3	Final approach below 5,000 ft RA with the VD on the ND.	An amber radio altitude zone under the aircraft symbol; the radio height is the same on both PFDs.

4. FADEC, EIU, engine idle and relight

> **Aircraft:** A32NX and A380X | **Where:** E/WD ENGINE indications, SD ENGINE page, thrust levers, ENG MASTER, flyPad Failures | **Status:** Sim-tested by the user in the coverage program group G1 (2026-10-18, cards A1-A5, B1-B7, F1, F3-F6 passed; the crank-without-FADEC bug it found was fixed). A32NX automatic TOGA after 8 s: sim-tested and OK (2026-10-25). Bleed status fault idle (HI GND IDLE) and A380X pack controllers lost (ENG IDLE HI): sim-tested OK on 2026-10-22. The 2026-10-12 to 2026-10-14 engine batches (P15, HP fuel valve, FADEC faults, derated take-off) were not sim-tested.

What it is for

FADEC power supply and channels (A32NX alternator and DC buses, A380X EEC channels and EIPMs), the engine interface units, idle modes (approach idle, higher idle with packs, HI GND IDLE with an ENG BLEED STATUS FAULT, ENG IDLE HI with both pack controllers lost), the A380X NORMAL / DEGRADED (N1) modes with the degraded accuracy indications, the A32NX automatic TOGA 8 s after the levers are set below TOGA without FLEX, automatic relight ignition, the N3 run-down and quick relight, the A380X engine overspeed failure, the FCOM EGT limits, REV, FLEX and N2 crank indications on the E/WD, and the pre-start motoring of a hot engine.

How to use

1. Nothing to set. Engine and FADEC failures are in flyPad > Failures (ATA 72, 73, 74, 77, 78, 79, 80).

How to test

#	Do this	Expected result
1	A32NX at take-off without a FLEX temperature: set the thrust levers between CL and TOGA and leave them.	ENG THR LEVERS NOT SET; 8 s later the FADEC selects TOGA thrust, until the levers go back to CL or below.
2	A380X on the ground, ENG MASTER 1 OFF at idle, then back ON after 10 s.	The N3 runs down slowly (about 45 % after 10 s); ENG MASTER ON relights the engine while the N3 shown is still above the relight limit.

3	A380X in flight: flyPad > Failures > ATA 73 > **Engine 1 overthrust (thrust control lost)**.	The FADEC shuts engine 1 down without automatic relight and ENG 1 FAIL shows the THRUST CTL LOST line.
4	A380X: fail both pack controllers (flyPad > Failures > ATA 21).	COND TEMP CTL FAULT, engines to idle high at idle thrust, STATUS INFO ENG IDLE HI.

5. FMS predictions and VNAV

> **Aircraft:** A32NX and A380X | **Where:** MCDU F-PLN, PERF, PROG, FUEL PRED, A380X MFD F-PLN, PERF, VERT REV, FUEL&LOAD, ND and VD | **Status:** Sim-tested by the user in the coverage program groups G1, G2 and G6 (2026-10-18 to 2026-10-23): energy circle, FLS beam, CRZ WIND page SAT / ALT, ALTN legs, SEC copy predictions, OPT FL with SAT entry: OK. Go-around and engine-out predictions and the 2026-10-12 / 13 MFD and MCDU page fixes (P8, P20, hold, EOSID, POSITION MONITOR): not sim-tested.

What it is for

The flight plan predictions and the VNAV logic follow the FCOM: ECON speeds and OPT FL from the cost index, weight and wind, the constant Mach segment, RTA time constraints with speed control, optimum step and VNAV messages, the energy circle and DES CABIN RATE, descent predictions with anti-ice, predictions on the alternate and secondary flight plans, go-around, take-off and engine-out predictions with drift down, SAT entries for the cruise, time markers as pseudo waypoints, MORE DRAG and SPD BRK messages, holds (database hold, last exit time, unfolded on the VD), the EOSID, the FM position monitor, FUEL PRED and PREDICTIVE GPS pages of the A32NX MCDU.

How to use

1. Enter the flight plan and the PERF data as usual; the predictions appear on the pages named above.

How to test

#	Do this	Expected result
1	A32NX with a flight plan, ZFW and cost index entered. Open PERF and PROG in cruise.	ECON cruise speed and OPT FL follow the cost index, weight and wind (not a fixed table and M.78).
2	A380X in descent toward the destination, ND in ARC or ROSE NAV.	A dashed energy circle within 180 NM of the destination; the VD shows the FLS beam on a non precision approach.
3	A380X MFD > F-PLN: add a hold with the HOLD revision; open the hold page and press DATABASE.	The hold becomes the database hold; LAST EXIT time and EFOB are predicted.
4	A380X: fail one engine in cruise (flyPad > Failures > ATA 72) and open PERF CRZ.	EO MAX, DRIFT DOWN TO FL and the EO CRZ MODE are shown.

6. A32NX GPWS: modes 1 and 2 rewritten, terrain ahead alerting (TAD), obstacle alerts and terrain clearance floor (TCF)

> **Aircraft:** A32NX | **Where:** Overhead GPWS panel (GPWS SYS, TERR, G/S MODE, FLAP MODE, LDG FLAP 3 pbs), GPWS/PULL UP lights on both glareshields, EMER CANC key, ND terrain display | **Status:** Tested in the sim by the user with the recorder (October 2026): take-off without false alert, FL100 toward Mount Fuji with the caution, the warning and AVOID TERRAIN; the ND TERR AHEAD message and threat areas tested the same way on 26 October 2026, and the AVOID TERRAIN voice the same day; the obstacle alerts at 3000 ft toward a 3125 ft tower near KFAR the same day (caution, warning, AVOID OBSTACLE); the obstacles in the ND terrain picture at Hillsboro (KVLY mast drawn red) on 28 October 2026. Not tested: the Himalaya at FL250, Innsbruck-type valley approaches. Unit tests for every part.

What it is for

Before, the A32NX GPWS only knew the basic modes, which work from the radio altimeter. Flying toward a mountain at FL100 gave no alert until the ground rose under the aircraft, far too late. The GPWS now also looks ahead: it knows the terrain around the aircraft from an elevation grid, and warns when your flight path and the terrain meet. Modes 1 and 2 were written again at the same time.

What you see and hear

Situation	You hear	You see
Terrain ahead, you can still avoid it (caution)	"TERRAIN AHEAD", repeated every 7 s	Amber GPWS light on both GPWS/PULL UP pbs; the terrain comes up on both NDs by itself, with an amber **TERR AHEAD** and the threat cells in yellow
Terrain right below the aircraft, too low for the place (terrain clearance floor)	"TOO LOW TERRAIN", repeated every 7 s	Amber GPWS light; the terrain does not come up by itself and there is no ND message
Terrain ahead, close (warning)	"TERRAIN AHEAD" once, then "PULL UP" again and again	Red PULL UP light; terrain on both NDs, red **TERR AHEAD**, the threat cells in red
Terrain ahead and the aircraft could not climb over it even with a pull-up	"AVOID TERRAIN" again and again (about every 1.5 s)	Red PULL UP light; terrain on both NDs, red **TERR AHEAD**, the warning cells in a red and black crosshatch
An obstacle (tower, mast) ahead, you can still avoid it (caution)	"OBSTACLE AHEAD", repeated every 7 s	Amber GPWS light; the terrain comes up on both NDs with an amber **OBST AHEAD** and the threat cells in yellow
An obstacle ahead, close (warning)	"OBSTACLE AHEAD" once, then "PULL UP" again and again	Red PULL UP light; red **OBST AHEAD**, the threat cells in red
An obstacle ahead that a pull-up could not clear	"AVOID OBSTACLE" again and again (about every 1.5 s)	Red PULL UP light; red **OBST AHEAD**, the warning cells in the red and black crosshatch
Mode 1 second boundary (sink rate too high near the ground), or mode 2 warning	"PULL UP"	Red PULL UP light
Any other basic alert (SINK RATE, mode 2 caution TERRAIN, DON'T SINK, TOO LOW ..., GLIDESLOPE)	The matching words	Amber GPWS light

- A warning takes over from a caution. Once AVOID TERRAIN has been said, PULL UP does not come back in the same warning.
- Terrain comes first: when the same scan finds terrain and an obstacle, you get the terrain alert. The obstacle words only come when the threat is an obstacle alone.
- The terrain display comes up on both NDs and the TERR ON ND pbs light, whatever they were set to. During the alert each pilot can switch it off with the TERR ON ND pb.
- **The AVOID TERRAIN voice:** the sound bank has no such word, so "AVOID" was made with a speech synthesiser converted to the GPWS voice and joined to the "TERRAIN" word of the sound bank (`tools/gpws-voice/generate.py` rebuilds it). The GPWS says one word at a time: the last PULL UP ends before the first AVOID TERRAIN, they never overlap. "AVOID OBSTACLE" is made the same way and waits for the last OBSTACLE AHEAD or PULL UP in the same way.

- **TERR STBY:** the predictive functions need the GPS position (accurate to 0.5 NM or better), the GPS altitude, the vertical speed and the terrain grid around the aircraft. When one is missing for 3 s the terrain function goes to standby (TERR STBY on the ND, no TAD or TCF alerts); 1 s after everything is back it works again. A block of the grid that is merely slow to load does not make it flicker.

On the ND: TERR AHEAD and the threat areas

- **OBST AHEAD** takes the place of TERR AHEAD, in the same colours and with the same flashing, when the alert is about an obstacle. The obstacle cells are drawn in the threat areas like terrain cells.
- **TERR AHEAD** on both NDs: amber for the caution TERRAIN AHEAD, red for the warning (TERRAIN AHEAD PULL UP or AVOID TERRAIN). TOO LOW TERRAIN has no ND message (FCOM: nil). It takes the place of the cyan **TERR** label above the terrain elevation box.
- It flashes for 9 s each time it is triggered (again when the caution becomes a warning), then stays steady while the alert lasts.
- If the terrain picture cannot be shown, the ND says why in its centre, red or amber with the alert: **TERR: CHANGE MODE** in PLAN mode, **TERR: REDUCE RANGE** at 160 and 320 NM.
- **Threat areas** are drawn over the terrain picture: the 1 arcminute cells (about 1 NM) that break the caution envelope in solid yellow, those that break the warning envelope in solid red, and in a red and black crosshatch while AVOID TERRAIN is on.
- They come from the look-ahead strip along your track, which is narrow and widens in turns. So they are small patches where your path meets the terrain, not whole mountains. Up to 64 runs of cells are drawn; when there are more, the nearest ones are kept.



Figure 1. Caution (ARC, 20 NM): amber TERR AHEAD at the lower right, the threat cells in yellow on the track.

In-sim screenshot taken during development: not representative of the final in-game UI, reference for the explanation only.



Figure 2. Warning: red TERR AHEAD, the warning cells in red next to the yellow caution cells.

In-sim screenshot taken during development: not representative of the final in-game UI, reference for the explanation only.



Figure 3. AVOID TERRAIN: the warning cells in the red and black crosshatch, yellow caution cells beside them.
In-sim screenshot taken during development: not representative of the final in-game UI, reference for the explanation only.

The functions in plain words

- **Terrain ahead alerting (TAD):** the GPWS scans the terrain along and beside your track, and checks whether you could clear it by levelling off and then climbing gently (caution), and whether you could still clear it if you pulled up now (warning). Both envelopes end on such a climb path, so the caution comes about a minute before the threat and the warning about half a minute before (the 3 degree climb of the caution path is a design choice; the manual gives no angle). It does not alert on the ground.
- **Obstacle alerts:** the same scan also reads an obstacle grid (the highest obstacle top in each cell) and gives the OBSTACLE words for it. No extra margin is added for obstacles, because their heights are surveyed values, not averaged terrain.
- **Terrain clearance floor (TCF):** around the destination runway (up to 15 NM) there is an invisible floor above the terrain. It starts low at the runway and rises with the distance. If you descend below it, you get TOO LOW TERRAIN whatever the gear and flaps are. The shape of the floor is rebuilt from the figures of the Honeywell guide; it is not a published curve.
- **Mountainous approach areas:** where the terrain within 6 NM of the runway is more than 2000 ft above it, the warning looks only 30 s ahead instead of 120 s, so that you are not warned on every approach there.
- **Landing tunnel:** when you are configured for landing (gear down, landing flaps), inside the approach sector of the runway (2.7 NM), in a tunnel above the approach path, and your track converges on the runway, the alerts are inhibited. That is how a normal landing near hills stays quiet.
- **Take-off:** no alerts on the ground; the terrain clearance floor does not work within 2.5 NM of the departure runway, and the scan starts afresh at lift-off.

The pushbuttons and keys

Control	What it does
TERR pb OFF (overhead)	Inhibits TAD and TCF. Mode 2 is then active again (it is switched off while the predictive functions work).

GPWS SYS pb OFF	Inhibits the basic modes 1 to 5 and their lights. TAD and TCF keep working: they do not need the radio altimeter or the airspeed.
EMER CANC	Silences the words of the alert that is going on (also the repeats), until that alert is over. The lights stay on. A new alert speaks again.
Audio inhibit	Stops the words of the predictive alerts too.

Modes 1 and 2 after the rewrite

- The lights follow the real format: **red PULL UP only for the second boundary of mode 1 and for a mode 2 warning** (and the hold after mode 2); every other alert is the amber GPWS light.
- Mode 2 has two nested envelopes: the outer gives the caution ("TERRAIN, TERRAIN"), the inner the warning ("PULL UP"). They depend on the radio altitude, its closure rate and the configuration.
- Mode 2 is inhibited while TAD and TCF work. It comes back when they fail, when TERR is OFF, or when the radio altitude and the height above the terrain computed from the grid differ by more than 2000 ft (one of the two is then wrong).
- The glideslope bias of mode 1 is applied only for a valid glideslope deviation. EMER CANC latches the silence until the alert is over.

Obstacles in the ND terrain picture

- With TERR ON ND selected, ARC and ROSE modes also draw the obstacles (towers, masts) from the obstacle file, in the same colours and dot density as the terrain at that height: an obstacle that is higher than you is red, one about at your level is amber, and so on. There is no separate symbol for an obstacle; this is how the A320 FCOM describes the picture.
- United States only: the obstacle file is the FAA Digital Obstacle File. Elsewhere, or without the file, the ND shows terrain only.
- Nothing is drawn in PLAN mode, as for the terrain.
- The peaks box (highest terrain figure) does not include obstacles: its figures come from SimBridge, which does not know them.

The escape manoeuvre

On a PULL UP or AVOID TERRAIN warning: autopilot off, pitch up firmly, TOGA thrust, wings level, and keep climbing until the alert stops (the usual Airbus terrain escape; the GPWS does not fly it for you).

Limitations and why

Limitation	Why	What to do
Obstacle alerts in the United States only.	The obstacle data is the FAA Digital Obstacle File, which lists the obstacles of the United States and its territories. Elsewhere there is no obstacle alert (and no TERR STBY for it: the terrain functions do not need the obstacle file).	Outside the United States, do not rely on it for man-made obstacles.
The obstacle data is a snapshot.	The grid is built once from the FAA file (26 October 2026); obstacles built or removed later are not in it until the file is rebuilt.	-
An obstacle alert can come a few seconds early.	An obstacle fills its whole 1 arcminute cell (about 1 NM), so the scan meets it up to about 1 NM before the real tower.	-

No runway database.	TCF, the mountainous approach area and the landing tunnel only know the runway threshold the FMGC sends for its destination and the place you took off from (the departure runway). An airport that is neither is not protected by a floor.	Enter the destination in the flight plan, with its runway when you know it.
Valley airports can alert.	The grid cells are 1 arcminute (about 1 NM), so a narrow valley airport such as Innsbruck sits in cells with the heights of the surrounding mountains. A margin that grows with the relief around the cell is added because the source data is averaged and reads peaks too low.	Expect TOO LOW TERRAIN or TERRAIN AHEAD there; set TERR OFF if you must.
Coverage from latitude -84 to 83.	The terrain source has nothing beyond that.	Beyond it: TERR STBY.
The grid is a file, and without it there is TERR STBY.	The grid is `panel/terrain_grid.bin` in the A32NX package folder (about 20 MB). It is too big for the repository, so it lives in the large-files repository (`TCdr/aircraft-large-files` , branch `feat/a32nx-taws-terrain-grid`). An aircraft without it has no predictive terrain alerts and shows TERR STBY.	Install a build that contains the file.
Threat areas only where the terrain picture is shown.	They are drawn over the terrain picture and cut to its area, so there are none in PLAN mode (TERR: CHANGE MODE instead).	Use ARC or ROSE NAV.
Threat areas are 1 arcminute cells.	That is the resolution of the terrain grid (about 1 NM): the patches are blocky and show the cells of the look-ahead strip, not the shape of the mountain.	-
AVOID TERRAIN is a made-up recording.	The sound bank has no such word: "AVOID" is synthesised and converted to the GPWS voice, so it can sound slightly different from the recorded words.	-
Position accuracy of 0.5 NM or better is needed.	Design choice: the accuracy at which the real system deactivates is not published. Half a cell was chosen so that the cell under the aircraft is the cell read.	-

Where the data comes from

The elevation grid is derived from the terrain data of SimBridge (the data behind the ND terrain display), reduced to one byte per 1 arcminute cell: 0 is water, otherwise the highest elevation of the cell in 50 m steps. It is built by `tools/taws-terrain-grid/build_grid.py`. The source is already averaged (Fuji reads about 3650 m, the real summit is 3776 m), which is why the alert logic adds its own margins. The aircraft reads the file from its own package while you fly; SimBridge does not have to run and nothing goes to the internet.

The obstacle grid `panel/obstacle_grid.bin` (about 265 KB, in the repository) has the same format: in each cell the highest obstacle top above sea level, rounded up to 50 m. It is built from the FAA Digital Obstacle File by `tools/taws-obstacle-grid/build_obstacle_grid.py`.

How to use

1. Nothing to set. Fly with the GPS working and a destination in the flight plan; the GPWS SYS and TERR pbs stay in.

2. Set TERR to OFF only to inhibit TAD and TCF (for instance on a deliberately low valley approach). Press EMER CANC to silence a repeating alert once you have taken note of it.

How to test

#	Do this	Expected result
1	A32NX in flight, GPS aligned, TERR and GPWS SYS pbs in. Look at the ND.	No TERR STBY. (Without `terrain_grid.bin` in the panel folder: TERR STBY.)
2	Take off normally from an airport near hills.	No alert on the ground or at lift-off.
3	Cruise at FL100 on a track that meets high ground (for example RJAA to RJOA, toward Mount Fuji).	About a minute before the terrain: "TOO LOW TERRAIN" then "TERRAIN AHEAD" every 7 s, amber GPWS light, terrain on both NDs.
4	Keep the course.	The warning: "TERRAIN AHEAD" once, "PULL UP" repeating, red PULL UP light. When a pull-up could not clear it: "AVOID TERRAIN", about every 1.5 s, never over a PULL UP.
5	Look at both NDs in ARC at 20 NM during the caution, then the warning, then AVOID TERRAIN.	Caution: amber TERR AHEAD in place of the cyan TERR label, flashing 9 s then steady, yellow cells on the track. Warning: TERR AHEAD red and flashing again, red cells. AVOID TERRAIN: the red cells crosshatched red and black.
6	During the alert, select PLAN mode, then ARC at 160 NM.	TERR: CHANGE MODE, then TERR: REDUCE RANGE, in the ND centre, in the colour of the alert; no threat areas.
7	Press EMER CANC during the alert.	The words stop and do not repeat for that alert; the lights stay.
8	Turn away and climb.	The alert ends, the lights go off. A new threat speaks again.
9	In the United States, fly at 3000 ft and 260 kt straight at a tall tower (for example from KFAR, DIR TO 4716.7N/09720.5W, a 3125 ft tower) and keep the level.	About 75 s out: "OBSTACLE AHEAD" every 7 s and amber OBST AHEAD; about 30 s out: "OBSTACLE AHEAD", "PULL UP", red OBST AHEAD; then "AVOID OBSTACLE" about every 1.5 s, never over a PULL UP.
10	Climb (AP off, pitch up, TOGA).	The obstacle alert ends.
11	Park the A32NX at Hillsboro, North Dakota (3H4). Power on, ND in ROSE NAV, range 20 NM, TERR ON ND.	A red patch about 9 NM west of the airport: the KVLY mast (about 2060 ft above the ground). Without `obstacle_grid.bin` in the panel folder the ND shows the terrain only. Nothing in PLAN mode.
12	Set the overhead TERR pb OFF in front of terrain.	No TAD or TCF alert; mode 2 works again from the radio altimeter.

7. Surveillance: GPWS test, TCAS traffic and failures, windshear and radio height voices, ILS side

> **Aircraft:** A32NX and A380X | **Where:** Overhead GPWS panel, EFIS CP TRAF pb, ND, MFD SURV page, PFD | **Status:** Committed in the FCOM gap program of 2026-10-12 to 2026-10-14 (batches P1 to P20 and Tier 2) without sim tests, on the decision that the community tests at pull request time. Unit tests only.

What it is for

Smaller surveillance items of the gap program: the A32NX GPWS can be tested on the ground with the PULL UP-GPWS pushbutton; the TRAFFIC selector of the THRT shows the proximate and other intruders on the ND; the A380X EFIS CP TRAF pb shows all the traffic and a TA or RA selects it; the sidestick priority voices (DUAL INPUT, PRIORITY LEFT / RIGHT), the intermediate radio height callouts and the WINDSHEAR voice; the A380X alpha floor inhibition above M 0.5; each PFD reads the ILS receiver of its own side; the MFD SURV status page shows a failed XPDR, TURB and PRED W/S failures and TAWS 1(2) IN USE.

How to use

1. Nothing to set. GPWS test: on the ground press and hold the PULL UP-GPWS test pushbutton.

How to test

#	Do this	Expected result
1	A32NX on the ground. Press the GPWS test pushbutton on the overhead panel.	GLIDESLOPE with the GPWS light, then PULL UP and TERRAIN TERRAIN PULL UP with the PULL UP light.
2	A380X with traffic around, EFIS CP TRAF pb off, then a TA appears.	The ND selects the traffic display by itself; the TRAF pb on shows all the traffic.
3	Move both sidesticks together on the ground with the A32NX powered.	DUAL INPUT voice every 5 s; a pilot holding priority for 1 s gives PRIORITY LEFT or PRIORITY RIGHT.

8. Flight control law fixes (ELAC, A380X protections)

> **Aircraft:** A32NX and A380X | **Where:** Sidestick, PFD speed scale, flyPad Failures (hydraulics) | **Status:** Sim-tested by the user on 2026-10-25: A32NX blue + yellow hydraulics lost keeps normal law, A32NX high speed protection, A380X 60 / 66 degree bank limit: OK. The A380X protection values were OK but the first version overshot to 389 kt with the stick held (A320 gains); the dive fix (VMO / MMO in a dive, 15 degree nose-down limit held) was not recorded as sim-tested.

What it is for

A32NX: high speed protection stops a full forward sidestick at VMO+16 kt / MMO+0.04 (not 380 kt / M0.88); the loss of blue and yellow hydraulics keeps pitch normal law; the sidestick takeover button must be held 40 s. A380X: A380 values for the high speed protection (active from VMO+10 kt, full forward holds VD-10 kt), bank limit 60 degrees with slats or flaps (67 clean), no nose-up limit drop in CONF FULL, direct law after the loss of the three IRs, abnormal attitude law from 30 degrees angle of attack, and the dive fix.

How to use

1. Nothing to set: fly the aircraft.

How to test

#	Do this	Expected result
1	A32NX in cruise. Lose the blue and yellow hydraulic systems (flyPad > Failures > ATA 29, for example the blue and yellow reservoir leaks).	The PFD stays in normal law for pitch (no alternate law message).
2	A32NX at level flight, push the sidestick fully forward and hold.	The speed stops rising at about VMO+16 kt.
3	A380X in a turn with slats extended, bank beyond 60 degrees.	The bank angle is limited to 60 degrees.

9. Aircraft system models: hydraulics, bleed, fuel, electrics, brakes, APU, ice, pressurization, ADIRS

> **Aircraft:** A32NX and A380X | **Where:** Overhead panels, SD pages, flyPad Failures | **Status:** Committed in the FCOM gap program of 2026-10-12 to 2026-10-14 (batches P1 to P20 and Tier 2) without sim tests, on the decision that the community tests at pull request time. Unit tests only.

What it is for

System models corrected or added from the FCOM gap audit, many with new flyPad failures: A380X brake pressures (5,000 psi system), ROP and RTO autobrake, gear speed and hydraulic supply (GREEN and YELLOW needed), hydraulic reservoir and electric pump modes, bleed overpressure valve, HP ground air, leak detection and crossbleed start; A380X fuel (load alleviation, AUTO GND XFR, trim tank feed, collector cells, XFR pushbuttons in AUTO); A32NX BSCU / ABCU, fuel temperature, wing tank overflow, APU APS3200 limits and automatic shutdowns, DC BUS 2 takeover after 5 s, DC ESS supply, RAT size, computer self-tests after a power outage, wing and engine anti-ice valves, cabin pressure without a landing elevation, ADIRS alignment limits, ADR / IR failures and power supplies, A380X electrical, braking and steering and ADR airspeed / AOA failures.

How to use

1. Nothing to set. The new failures are in flyPad > Failures under their ATA chapter.

How to test

#	Do this	Expected result
1	A380X on the ground with one electric hydraulic pump only.	The pressure rises steadily to normal (no saw-tooth) while the drooped flight controls re-centre.
2	A380X: fail the GREEN hydraulic system in flight, then lower the gear.	The main gears do not extend normally (GREEN and YELLOW are both needed; the nose gear needs GREEN).
3	A32NX in flight: flyPad > Failures > ATA 24 > **DC 1**, Fail.	DC BUS 2 takes over DC BUS 1 and the DC BAT BUS after 5 s.
4	A32NX cold and dark, APU running: cause an APU ECB failure (flyPad > Failures > ATA 49).	The APU shuts down automatically with a FAULT.

10. Panels, communications, effects, flyPad and GSX

> **Aircraft:** A32NX and A380X | **Where:** RMP, CALLS panel, overhead panels, flyPad Ground and Settings, outside view | **Status:** Committed in the FCOM gap program of 2026-10-12 to 2026-10-14 (batches P1 to P20 and Tier 2) without sim tests, on the decision that the community tests at pull request time. Unit tests only. The GSX power sync re-read was part of the GSX power sync work (2026-09-30).

What it is for

Cockpit panel and flyPad items of the program: the A380X STBY RAD NAV key, held PA key, RMP message queue, CALLS panel (each button calls its own station) and ELT selector, the loudspeaker volume knobs, the A32NX CVR automatic power and CVR TEST, COMPANY COM inbox with the received AOC messages, engine and APU fire, engine damage and oil leak effects on both aircraft (Settings > Sim Options > Situation Visual Effects), the ground jettison drip and contrails per engine, the visible A32NX RAT, the exhaust haze per engine, the A380 taxi width check and taxi route with real ATC clearances, the A380 Services door buttons, pushback map performance, external power following the sim ground power and the GSX power sync setting.

How to use

1. Effects: flyPad > Settings > Sim Options, switch **Situation Visual Effects** on or off (on by default).
2. The other items need no setting.

How to test

#	Do this	Expected result
1	A32NX, engine 1 fire (flyPad > Failures > ATA 26 > **Engine 1 fire**), outside view.	Flames at the exhaust of engine 1 until the fire is out (nothing during a FIRE TEST).
2	A380X, CALLS panel: press PILOT REST.	The cabin does not answer CABIN READY; the EMER ON and CALL lights come on steady.
3	A32NX cold and dark, battery on: wait, then start an engine.	The CVR runs by itself during the first 5 min, with an engine running and in flight.